Ex No: 10 Performance Optimization in Zoho Creator

Pragadeesh - 240701388

1. Aim

To demonstrate performance optimization techniques in the Patient Appointment application, mainly efficient record retrieval and efficient workflow processing.

2. Procedure

1. Open the workflow *Send_Appointment_Acknowledgement* from Experiment 7.
2. Observe the efficient pattern, which fetches only the record that was just submitted using a criteria:

```
appt = Appointment_Booking_Form[ID == input.ID];
if(appt.Email_Address.isBlank())
{
    info "No email - skipped";
}
else
{
    sendmail [ from: zoho.adminuserid to: appt.Email_Address subject:
"Appointment received" message: "Thank you." ]
}
```

3. Compare with the inefficient pattern, shown only for illustration and not used in the application:

```
allAppts = Appointment_Booking_Form[];
for each rec in allAppts
{
    if(rec.ID == input.ID)
    {
        // work on rec
    }
}
```

4. Note the techniques applied across the application:
    - Fetch with criteria rather than loading all records.
    - Use `input.ID` to identify the current record.

- Check blank values before sending email.
- Process large lists with Batch Workflows (150 records per batch) instead of one long loop.
- Restrict the report workflow to fire only when Consultation Fee or Insurance Coverage Percent changes.

5. Run both patterns on the sample data and compare execution time in the logs.

[Insert screenshot: Efficient workflow script]

[Insert screenshot: Execution log comparison]

3. Result

Criteria-based fetching and batch processing reduced unnecessary data retrieval, which keeps the appointment application responsive as the number of bookings grows.